

Fit a Line Answers

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In [1]: %matplotlib ipynb
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
```

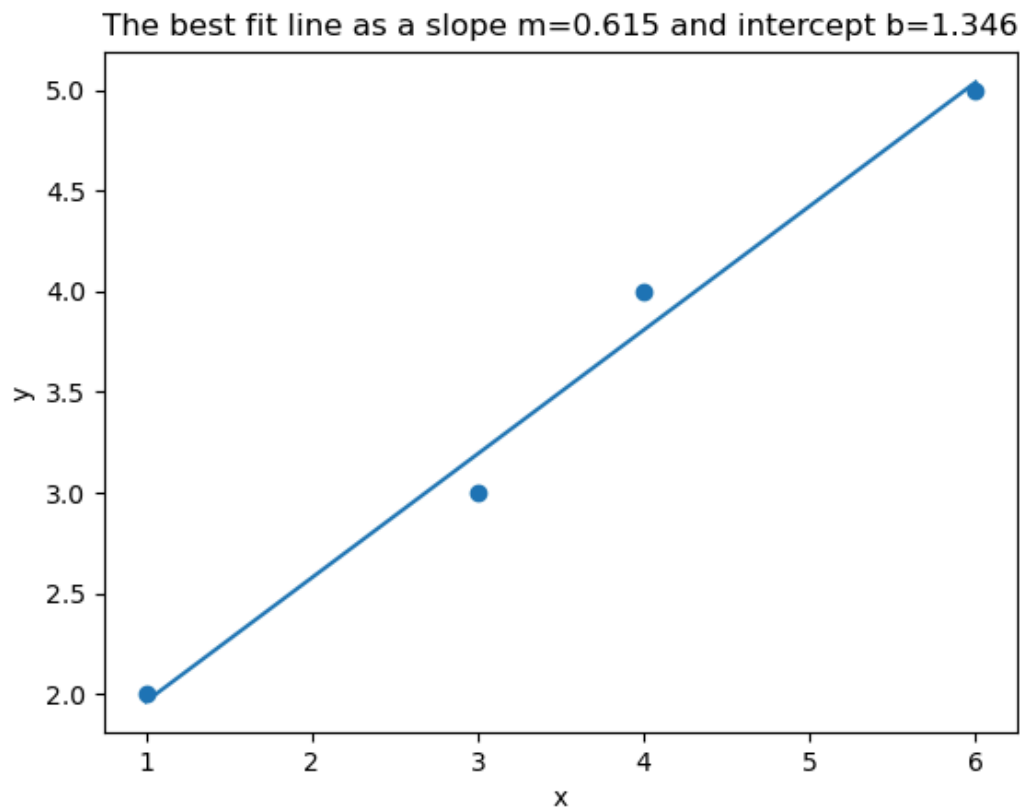
```
In [2]: x=np.array([1.,3,4,6])
y=np.array([2,3,4,5])

xy=x*y
n=len(x)
m=(n*sum(xy)-sum(x)*sum(y))/(n*sum(x*x)-sum(x)**2)
b=(sum(x*x)*sum(y)-sum(xy)*sum(x))/(n*sum(x*x)-sum(x)**2)
x_fit=np.linspace(min(x),max(x))
y_fit=m*x_fit+b

fig,ax=plt.subplots()
ax.scatter(x,y)
ax.plot(x_fit,y_fit)
ax.set_xlabel('x')
ax.set_ylabel('y')
title='The best fit line as a slope m={:.3f} and intercept b={:.3f}'.format(m,b)
ax.set_title(title)
```

Out[2]: Text(0.5, 1.0, 'The best fit line as a slope m=0.615 and intercept b=1.346')

Figure



```

In [3]: KLGA=np.array([52,43,39,34,33,31,34])
        MONT=np.array([53,44,40,37,36,30,32])

        x=KLGA
        y=MONT
        xy=x*y
        n=len(x)
        m=(n*sum(xy)-sum(x)*sum(y))/(n*sum(x*x)-sum(x)**2)
        b=(sum(x*x)*sum(y)-sum(xy)*sum(x))/(n*sum(x*x)-sum(x)**2)
        x_fit=np.linspace(min(x),max(x))
        y_fit=m*x_fit+b

        fig,ax=plt.subplots()
        ax.scatter(x,y)
        ax.plot(x_fit,y_fit)
        ax.set_xlabel('KLGA')
        ax.set_ylabel('MONT')

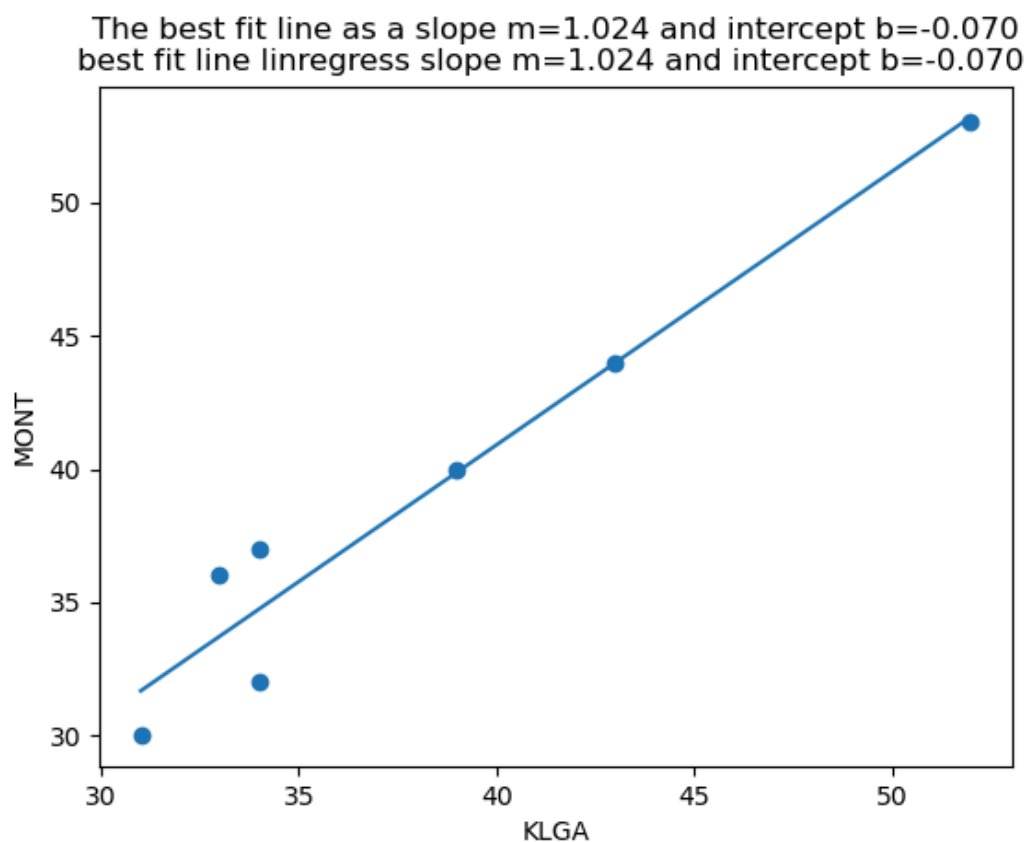
        linregress_out=stats.linregress(x,y)

        title=('The best fit line as a slope m={:.3f} and intercept b={:.3f}'.format(m,b)+
              '\nbest fit line linregress slope m={:.3f} and intercept b={:.3f} '\
              .format(linregress_out[0],linregress_out[1]))
        ax.set_title(title)

```

Out[3]: Text(0.5, 1.0, 'The best fit line as a slope m=1.024 and intercept b=-0.070\nbest fit line linregress slope m=1.024 and intercept b=-0.070 ')

Figure



Now that is only a decent plot. We don't have units on the axis and I will show you how to represent the m, b, r^2 and p on the graph. Here we go.

```

In [4]: KLGA=np.array([52,43,39,34,33,31,34])
        MONT=np.array([53,44,40,37,36,30,32])

        fig,ax=plt.subplots()
        fig.set_size_inches(6,6)
        ax.scatter(KLGA,MONT,s=50)

        ax.set_xlabel('KLGA ($^{\circ}$F)',fontsize=14)
        ax.set_ylabel('MONT ($^{\circ}$F)',fontsize=14)

        ax.tick_params(axis='both', which='major', labelsize=14) #This increases the font si

        linregress_out=stats.linregress(KLGA,MONT)

        x_fit=np.linspace(min(KLGA),max(KLGA))
        y_fit=linregress_out[0]*x_fit+linregress_out[1]
        ax.plot(x_fit,y_fit)

        props=dict(boxstyle='round',facecolor='green',alpha=0.5)

        textstr='m={:.3f}\nb={:.3f}\n$r^2$={:.3f}\n$p$={:.3f}'\
                .format(linregress_out[0],linregress_out[1]\
                        ,linregress_out[2]**2,linregress_out[3])

        ax.text(0.05,0.95,textstr,transform=ax.transAxes,fontsize=10\
                ,verticalalignment='top',bbox=props)

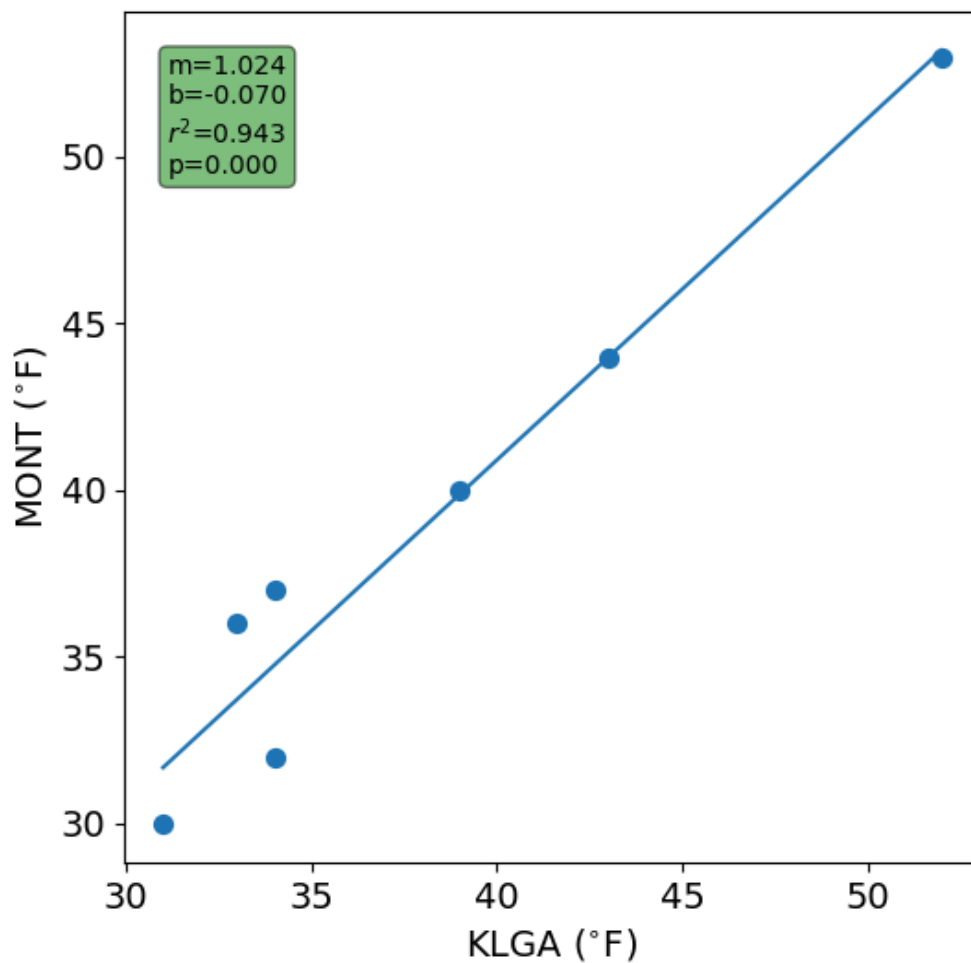
```

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Out[4]: Text(0.05, 0.95, 'm=1.024\nb=-0.070\n$r^2$=0.943\n$p$=0.000')

```

Figure



Now we are starting to look professional!!! and can make a figure caption. Figure 1. Temperature at JFK Airport (KNYC) versus temperature at Laguardia Airport (KLGA). The line is the best fit linear regression.

In []:

In []: